LOG-CONCAVITY OF INDEPENDENCE POLYNOMIALS OF CATERPILLARS

KNOWLEDGE SENSE

Abstract. A caterpillar is a tree whose non-leaf vertices induce a path, possibly with one vertex or none. We prove that the independence polynomial of every caterpillar is log-concave, for arbitrary numbers of pendant vertices along the path. Using the symmetric function criterion of Li, Li, Yang and Zhang, we show that $I(T;s)\,I(T;t)$ is Schur positive in two variables, by a transfer-matrix argument that reduces this product to free Ising chains after an explicit contraction step.

1. Introduction

For a graph G let $i_k(G)$ be the number of independent sets of G with k vertices, and let

$$I(G;x) = \sum_{k\geq 0} i_k(G)\, x^k$$

be the independence polynomial of G [6]. Alavi, Malde, Schwenk and Erdős [1] showed that the numbers $i_1(G), \ldots, i_{\alpha(G)}(G)$ can be ordered in any prescribed way, and asked whether the independence sequence is unimodal when G is a tree or a forest [1, Problem 3, p. 21]. A sequence (a_k) is log-concave if $a_k^2 \geq a_{k-1}a_{k+1}$ for all $k \geq 1$; for nonnegative sequences without internal zeros this implies unimodality. Levit and Mandrescu conjectured that the independence polynomial of any (well-covered) forest is log-concave [9, Section 3]. For general trees this fails: Kadrawi, Levit, Yosef and Mizrachi [8] found two trees on 26 vertices whose independence sequences are not log-concave, Kadrawi and Levit [7] constructed infinite families of such trees, and Galvin [4] showed that log-concavity can fail arbitrarily far from the end of the sequence. Thus log-concavity can hold only for restricted classes of trees.

A *caterpillar* is a tree whose non-leaf vertices induce a path, possibly with one vertex or none. For $n \geq 1$ and integers $m_1, \ldots, m_n \geq 0$ let $T(m_1, \ldots, m_n)$ be the tree obtained from a path with vertices $1, \ldots, n$ (the *spine*) by attaching m_i pendant vertices to vertex i for each i. Every caterpillar is of this form, and every tree of this form is a caterpillar; paths and stars are the two extreme cases. Wang and Zhu [13, Proposition 3.1] proved that the regular caterpillars $T(m, \ldots, m)$ have log-concave independence polynomials. Galvin and Hilyard [5, Theorem 1.9(1)] proved log-concavity for $T(\ell_1, \ell_2, \ldots, \ell_1, \ell_2)$ with an even number of spine vertices and arbitrary ℓ_1, ℓ_2. They also proved that attaching sufficiently many pendant edges

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at a fixed vertex of an arbitrary graph yields a log-concave independence polynomial [5, Theorem 1.11], and deduced that $T(k_1, \ldots, k_n)$ has a unimodal independence sequence whenever k_n is sufficiently large in terms of $k_1, \ldots, k_{n-1}$ [5, Corollary 1.12]. Bahls, Ethridge and Szabo [2, Theorem 1.1] proved unimodality when the numbers of pendant vertices are nondecreasing along the spine and satisfy further growth conditions. In the January 2026 version of [4], unimodality was recorded as open even for caterpillars.

Li, Li, Yang and Zhang [10] introduced a method based on chromatic symmetric functions and proved that every spider, that is, every tree with at most one vertex of degree at least three, has a log-concave independence polynomial [10, Theorem 3.1]. Caterpillars may have arbitrarily many vertices of degree at least three, while the spider with three legs of length two is not a caterpillar, so neither class contains the other.

Our main result is the following.

Theorem 1.1. *Let $n \geq 1$ and $m_1, \ldots, m_n \geq 0$, and let $T = T(m_1, \ldots, m_n)$. Then*

$$i_k(T)^2 \;\geq\; i_{k-1}(T)\, i_{k+1}(T) \qquad \text{for every } k \geq 1.$$

Equivalently, the independence polynomial of every caterpillar is log-concave.

Here $i_k(T) = 0$ for $k < 0$ and $k > \alpha(T)$, so the inequality is only of interest for $1 \leq k < \alpha(T)$. To our knowledge, log-concavity of the independence polynomials of all caterpillars has not appeared in the literature.

Theorem 1.1 implies that every caterpillar has a unimodal independence sequence. This consequence is also contained in the claim of Li, Vallier and Zhang [11], made in a provisional manuscript that builds on earlier manuscripts of Zhang and Li, that every finite forest has a unimodal independence sequence. Fang, Lu, Nevo, Yao and Zheng [3] proved unimodality for all sufficiently large forests, together with strict log-concavity in the range $n/5 \leq k \leq 17\alpha(F)/25$ for forests F on n vertices with n sufficiently large. Neither result gives log-concavity at every rank, and our argument is independent of both. Theorem 1.1 does not extend to real-rootedness or to vertex-weighted counts (Remarks 6.9 and 6.10).

Method. Stanley [12, Theorem 2.11] showed that a real polynomial P with $P(0) = 1$ has only negative real zeros if and only if the symmetric function $\prod_i P(x_i)$ is Schur positive. Li, Li, Yang and Zhang [10, Theorem 1.3] proved the analogous criterion for log-concavity, in terms of the Schur functions with at most two parts. In two variables s, t it says that $P(s)P(t)$ is a nonnegative combination of the polynomials $(st)^b h_a(s,t)$, where h_a is the complete homogeneous polynomial (Lemma 2.2). We prove this positivity for $P = I(T;\cdot)$ by a transfer-matrix argument along the spine.

After elementary reductions (Section 3), the proof reduces to the Schur positivity of a family of three-state transfer-matrix sums $Z_{\mathbf{j}}$ indexed by words $\mathbf{j}$ of nonnegative integers (Theorem 3.5). Each $Z_{\mathbf{j}}$ is a weighted sum over colorings of a sequence of charges, and the multiplicity of a coloring is affine in each of the indicators that consecutive charges have equal colors. A sign identity rewrites it as a combination of free Ising chains with couplings 0 and 2, which are Schur positive (Lemma 2.4). The coefficients of this combination can be negative. They arise only at spine vertices with exactly three pendant vertices in the union of the two independent sets, after a negative sign has been assigned to the pair of charges of such a vertex. Summing

out the two opposite colorings of this pair replaces the vertex by the matrix

$$B = \begin{pmatrix} 6 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & 0 \end{pmatrix}$$

in the transfer product (Proposition 5.1). The products of B with powers of the compatibility matrix C satisfy explicit inequalities (Lemma 5.3). With these we show that the occupied-to-vacant ratio of a partial transfer product stays in the interval $[-1/2, 1]$ along the spine, which gives the nonnegativity of the remaining coefficients (Proposition 6.7).

Organization. Section 2 collects facts on symmetric polynomials in two variables and proves the Schur positivity of free Ising chains. Section 3 reduces Theorem 1.1 to Theorem 3.5. Section 4 sets up the transfer matrices and the charges. Section 5 performs the contraction of negatively signed pairs and bounds the resulting kernels. Section 6 carries out the sign expansion and the boundary-ratio argument, and completes the proofs.

2. Preliminaries

2.1. **Symmetric polynomials in two variables.** Let Λ be the real vector space of symmetric polynomials in two variables s, t, and put

$$u = s + t, \qquad v = st, \qquad h_a = \sum_{i=0}^{a} s^i t^{a-i} \quad (a \geq 0).$$

For a partition $(a+b, b)$ the Schur polynomial in two variables is $s_{(a+b,b)}(s,t) = v^b h_a$, and Schur polynomials of partitions with at most two parts form a basis of Λ; hence so do the polynomials $v^b h_a$ with $a, b \geq 0$. Let $\Lambda^+ \subset \Lambda$ be the set of nonnegative linear combinations of these polynomials. An element of Λ^+ is called *Schur positive*.

Lemma 2.1. *The following hold.*

(a) *Let $f = \sum_{a,b} q_{a,b} s^a t^b \in \Lambda$. For $a \geq b \geq 0$ the coefficient of $v^b h_{a-b}$ in f equals $q_{a,b} - q_{a+1,b-1}$, where $q_{a+1,-1} = 0$.*

(b) *For $a, b \geq 0$ we have $h_a h_b = \sum_{r=0}^{\min(a,b)} v^r h_{a+b-2r}$.*

(c) *Λ^+ is closed under addition, multiplication and multiplication by nonnegative reals. In particular u^k, v^k and $(u+v)^k$ belong to Λ^+ for all $k \geq 0$.*

Proof. (a) Write $f = \sum c_{a',b'} v^{b'} h_{a'-b'}$, the sum over $a' \geq b' \geq 0$. Since

$$v^{b'} h_{a'-b'} = \sum_{i=b'}^{a'} s^i t^{a'+b'-i},$$

the monomial $s^a t^b$ with $a \geq b$ occurs in $v^{b'} h_{a'-b'}$ if and only if $a' + b' = a + b$ and $a' \geq a$, and then with coefficient 1. Hence $q_{a,b} = \sum_{r \geq 0} c_{a+r,b-r}$, and subtracting the same identity for $(a+1, b-1)$ gives the claim.

(b) For $a, b \geq 1$, multiplying by $(s-t)^2$ and using $(s-t)h_a = s^{a+1} - t^{a+1}$ shows that

$$(s-t)^2\big(h_a h_b - v\, h_{a-1} h_{b-1}\big) = s^{a+b+2} + t^{a+b+2} - s^{a+b+1}t - st^{a+b+1} = (s-t)^2 h_{a+b},$$

so $h_a h_b = h_{a+b} + v\, h_{a-1} h_{b-1}$. Since $h_0 = 1$, induction on $\min(a, b)$ gives (b).

(c) Closure under products follows from (b), and $u = h_1$, $v = v\, h_0$. □

The next lemma is the two-variable form of the criterion of Li, Li, Yang and Zhang. For completeness we include the short proof.

Lemma 2.2 (cf. [10, Theorem 1.3])**.** *Let $P(x) = \sum_k a_k x^k$ be a polynomial with real coefficients, and put $a_k = 0$ for $k < 0$. If $P(s)P(t) \in \Lambda^+$, then $a_k^2 \geq a_{k-1}a_{k+1}$ for every $k \geq 1$.*

Proof. Write $P(s)P(t) = \sum_{a,b} q_{a,b} s^a t^b$. Then $q_{k,k} = a_k^2$ and $q_{k+1,k-1} = a_{k+1}a_{k-1}$, so by Lemma 2.1(a) the coefficient of $v^k h_0$ is $a_k^2 - a_{k+1}a_{k-1}$, which is nonnegative by assumption. □

Remark 2.3. For $P(0) = 1$, Li, Li, Yang and Zhang [10, Theorem 1.3], building on Stanley [12, Theorem 2.11], consider $F_P = \prod_{i \geq 1} P(x_i)$ and prove that, when all coefficients of P are positive, the following are equivalent: the coefficient of s_λ in F_P is nonnegative for all partitions λ with at most two parts; P is log-concave; P is strongly log-concave. Setting all variables except $x_1 = s$ and $x_2 = t$ equal to zero maps F_P to $P(s)P(t)$, annihilates the Schur functions with at least three parts and maps $s_{(a+b,b)}$ to $v^b h_a$. Hence the first condition is equivalent to $P(s)P(t) \in \Lambda^+$. We use only the implication of Lemma 2.2.

2.2. **Free Ising chains.** Throughout, we also use s and t as the names of two *colors*. For $\lambda \geq 0$ and colors $\kappa, \kappa' \in \{s, t\}$ put

$$W_\lambda(\kappa, \kappa') = \begin{cases} \lambda, & \kappa = \kappa', \\ 1, & \kappa \neq \kappa'. \end{cases}$$

For $N \geq 1$ and $\boldsymbol{\lambda} = (\lambda_1, \ldots, \lambda_{N-1}) \in [0, 2]^{N-1}$ define the Ising chain with free boundary conditions

$$X_N(\boldsymbol{\lambda}) = \sum_{\kappa \in \{s,t\}^N} \kappa_1 \kappa_2 \cdots \kappa_N \prod_{e=1}^{N-1} W_{\lambda_e}(\kappa_e, \kappa_{e+1}) \in \Lambda,$$

where the product $\kappa_1 \cdots \kappa_N$ is a monomial in s, t. Put $X_0 = 1$.

Lemma 2.4. *For every $N \geq 1$ and every $\boldsymbol{\lambda} \in [0, 2]^{N-1}$ we have $X_N(\boldsymbol{\lambda}) \in \Lambda^+$. Also $X_0 = 1 \in \Lambda^+$.*

Proof. Step 1: all couplings equal to 2. Let $X_N = X_N(2, \ldots, 2)$ for $N \geq 1$. With $\Delta = \operatorname{diag}(s, t)$ and $\mathbf{1} = (1, 1)^{\mathsf{T}}$ we have $X_N = \mathbf{1}^{\mathsf{T}} \Delta (W_2 \Delta)^{N-1} \mathbf{1}$, where W_2 also denotes the matrix $\left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$. The matrix $W_2\Delta = \left(\begin{smallmatrix} 2s & t \\ s & 2t \end{smallmatrix}\right)$ has trace $2u$ and determinant $3v$, so the Cayley–Hamilton theorem gives $X_N = 2uX_{N-1} - 3vX_{N-2}$ for $N \geq 3$, while $X_1 = u$ and $X_2 = 2s^2 + 2st + 2t^2 = 2u^2 - 2v$. Comparing coefficients of powers of a formal variable z, it follows that

$$\bigl(1 - 2uz + 3vz^2\bigr)\Bigl(1 + 2\sum_{N \geq 1} X_N z^N\Bigr) = 1 - vz^2.$$

Let $D = (1 - 2sz)(1 - 2tz) = 1 - 2uz + 4vz^2$. Then $1 - 2uz + 3vz^2 = D - vz^2$, and in the ring of formal power series in z with coefficients in Λ,

$$\frac{1 - vz^2}{D - vz^2} = (1 - vz^2) \sum_{r \geq 0} \frac{(vz^2)^r}{D^{r+1}} = \frac{1}{D} + \sum_{r \geq 1} (vz^2)^r \, \frac{1}{D^r}\Bigl(\frac{1}{D} - 1\Bigr).$$

Since

$$D^{-1} = \sum_{a \geq 0} 2^a h_a z^a,$$

every coefficient of D^{-1}, of $D^{-1} - 1$ and, by Lemma 2.1(c), of D^{-r} lies in Λ^+. Each power of z receives finitely many contributions, so $X_N \in \Lambda^+$ for all N.

Step 2: couplings in $[1, 2]$. For $\lambda \in [1, 2]$ we have $W_\lambda = (\lambda - 1)W_2 + (2 - \lambda)W_1$. Expanding the product over the edges, $X_N(\boldsymbol{\lambda})$ becomes a nonnegative combination of chains whose couplings are 1 or 2. Since $W_1(\kappa, \kappa') = 1$, a coupling 1 splits the chain into two independent chains, so each such chain is a product of chains with all couplings equal to 2 and lies in Λ^+ by Step 1 and Lemma 2.1(c).

Step 3: the general case. We use induction on $N \geq 1$; the case $N = 1$ is trivial since $X_1 = u$, and Step 2 covers the case that all couplings are at least 1. Suppose $\lambda_e < 1$ for some e. Since $W_{\lambda_e} = \lambda_e W_1 + (1 - \lambda_e)W_0$, the chain $X_N(\boldsymbol{\lambda})$ is a nonnegative combination of two terms. In the first, the edge e has coupling 1 and the chain splits into a product of two shorter chains, which lie in Λ^+ by induction (a chain with one charge is $X_1 = u$). In the second, $W_0(\kappa_e, \kappa_{e+1})$ is the indicator of $\kappa_e \neq \kappa_{e+1}$, so the two charges $e, e + 1$ contribute the factor $\kappa_e \kappa_{e+1} = v$. We sum over the two admissible pairs (κ_e, κ_{e+1}). If $1 < e < N - 1$, let $a = \lambda_{e-1}$ and $b = \lambda_{e+1}$; writing W_0 also for the matrix $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$,

$$W_a W_0 W_b = \begin{pmatrix} a + b & 1 + ab \\ 1 + ab & a + b \end{pmatrix} = (1 + ab)\, W_{(a+b)/(1+ab)},$$

so the second term equals $v(1 + ab)$ times a chain on $N - 2$ charges in which the coupling between charges $e - 1$ and $e + 2$ is $(a + b)/(1 + ab)$. This number lies in $[0, 2]$, because the function $2ab + 2 - a - b$ is affine in each variable and takes the values $2, 0, 0, 6$ at the corners of $[0, 2]^2$. If $e = 1 < N - 1$, the second term is $v \sum_{\kappa_2} W_b(\kappa_2, \kappa_3)$ times a chain on the charges $3, \ldots, N$, that is, $v(1 + b)X_{N-2}(\lambda_3, \ldots, \lambda_{N-1})$; the case $e = N - 1 > 1$ is symmetric. If $N = 2$, the second term is $2v = 2vX_0$. In all cases the second term lies in Λ^+ by induction and Lemma 2.1(c). □

3. Reduction to a normalized model

A *word* is a finite sequence $\mathbf{j} = (j_1, \ldots, j_n)$ of nonnegative integers, $n \geq 0$. All models below are sums over the assignments of a state to each site $i = 1, \ldots, n$, with free ends, in which the states of consecutive sites must be compatible and the state of site i receives a weight depending on j_i; the empty word gives the value 1. Put $U = u + v = s + t + st$.

Definition 3.1. We use the following four models.

(1) The *four-state model* $A_{\mathbf{j}}$ has states $(\beta, \beta') \in \{0, 1\}^2$, where (β_1, β_1') and (β_2, β_2') are compatible if $\beta_1 \beta_2 = 0$ and $\beta_1' \beta_2' = 0$, and weights

$$\omega_j(0, 0) = U^j, \quad \omega_j(1, 0) = st^j, \quad \omega_j(0, 1) = ts^j, \quad \omega_j(1, 1) = \begin{cases} v, & j = 0, \\ 0, & j \geq 1. \end{cases}$$

(2) The *three-state model* $G_{\mathbf{j}}$ is the restriction of $A_{\mathbf{j}}$ to the three states $(0, 0)$, $(1, 0)$ and $(0, 1)$, which we denote by 0, s and t, respectively. Two of these states are compatible unless both equal s or both equal t.

(3) The model $J_{\mathbf{j}}$ has the same states and compatibility as $G_{\mathbf{j}}$. For $j \geq 1$ the states 0, s, t have weights jvu^{j-1}, st^j, ts^j; for $j = 0$ they have weights 0, s, t.
(4) The *normalized model* $Z_{\mathbf{j}}$ has the same states and compatibility as $G_{\mathbf{j}}$. For $j \geq 1$ the states 0, s, t have weights ju^{j-1}, t^{j-1}, s^{j-1}; for $j = 0$ they have weights 0, s, t.

Lemma 3.2. *For $T = T(m_1, \dots, m_n)$,*

$$I(T; s)\, I(T; t) = \sum_{0 \leq j_i \leq m_i} \prod_{i=1}^{n} \binom{m_i}{j_i} A_{(j_1,\dots,j_n)}(s, t).$$

Proof. The left side is $\sum s^{|I_1|} t^{|I_2|}$ over pairs (I_1, I_2) of independent sets of T. Let L_i be the set of pendant vertices at the spine vertex i. Group the pairs according to the sets $Y_i = (I_1 \cup I_2) \cap L_i$; there are $\prod_i \binom{m_i}{j_i}$ choices of the sets Y_i with $|Y_i| = j_i$. For fixed Y_i, record at each spine vertex i the state (β_i, β_i'), where $\beta_i = 1$ if $i \in I_1$ and $\beta_i' = 1$ if $i \in I_2$. Independence of I_1 and I_2 on the spine is exactly the compatibility of consecutive states. A vertex of Y_i lies in I_1 only, in I_2 only, or in both, subject to the condition that it is not in I_1 if $\beta_i = 1$ and not in I_2 if $\beta_i' = 1$. Summing over these choices and including the factor $s^{\beta_i} t^{\beta_i'}$ of the spine vertex gives the weight $\omega_{j_i}(\beta_i, \beta_i')$. □

Lemma 3.3. *If $G_{\mathbf{j}'} \in \Lambda^+$ for every word $\mathbf{j}'$, then $A_{\mathbf{j}} \in \Lambda^+$ for every word $\mathbf{j}$.*

Proof. Fix the set D of sites in state $(1, 1)$. Its weight vanishes unless $j_i = 0$ for all $i \in D$; the set D contains no two consecutive sites, and the neighbors of D are forced into the state $(0, 0)$, which is compatible with every state. Hence

$$A_{\mathbf{j}} = \sum_{D} v^{|D|} \prod_{i \in \partial D} U^{j_i} \prod_{\Gamma} G_{\mathbf{j}|_\Gamma},$$

where D runs over sets of pairwise nonconsecutive sites with $j_i = 0$ on D, ∂D is the set of sites outside D adjacent to D, and Γ runs over the maximal intervals of sites outside $D \cup \partial D$. Every factor lies in Λ^+. □

Lemma 3.4. *For every word $\mathbf{j}$ we have $J_{\mathbf{j}} = v^{\nu(\mathbf{j})} Z_{\mathbf{j}}$, where $\nu(\mathbf{j})$ is the number of indices with $j_i \geq 1$. If $Z_{\mathbf{j}'} \in \Lambda^+$ for every word $\mathbf{j}'$, then $G_{\mathbf{j}} \in \Lambda^+$ for every word $\mathbf{j}$.*

Proof. For $j \geq 1$ the weights of J are v times those of Z, since $st^j = v\, t^{j-1}$ and $ts^j = v\, s^{j-1}$; for $j = 0$ they coincide. This proves the first statement. For the second, write the weight of the state 0 in G as $U^j = \omega_j + \varrho_j$, where $\omega_j = jvu^{j-1}$ and

$$\varrho_j = U^j - jvu^{j-1} = u^j + \sum_{d=2}^{j} \binom{j}{d} v^d u^{j-d} \qquad (j \geq 1),$$

and $\omega_0 = 0$, $\varrho_0 = 1$. Expanding the product of these sums, we choose a set R of sites that are in state 0 with weight ϱ_{j_i}. Since the state 0 is compatible with every state, the remaining sites split into maximal intervals Γ, each contributing $J_{\mathbf{j}|_\Gamma}$. Thus $G_{\mathbf{j}} = \sum_R \prod_{i \in R} \varrho_{j_i} \prod_\Gamma J_{\mathbf{j}|_\Gamma}$, and every factor lies in Λ^+ by Lemma 2.1(c) and the first statement. □

The remaining sections prove the following theorem.

Theorem 3.5. *For every word $\mathbf{j}$ we have $Z_{\mathbf{j}} \in \Lambda^+$.*

4. Transfer matrices and charges

We index 3×3 matrices and vectors by the states $0, s, t$, in this order, write $E_{\sigma\tau}$ for the matrix units and $\mathbf{1} = (1,1,1)^{\mathsf{T}}$, and let

$$C = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$$

be the compatibility matrix. With $\Omega_j = \operatorname{diag}(ju^{j-1}, t^{j-1}, s^{j-1})$ for $j \geq 1$ and $\Omega_0 = \operatorname{diag}(0, s, t)$, we have $Z_{\mathbf{j}} = \mathbf{1}^{\mathsf{T}}\Omega_{j_1} C\, \Omega_{j_2} C \cdots C\, \Omega_{j_n}\mathbf{1}$ for $n \geq 1$. Note that Ω_1 is the identity matrix.

Lemma 4.1. *For $\ell \geq 1$,*

$$C^\ell = \begin{pmatrix} d & c & c \\ c & a & b \\ c & b & a \end{pmatrix}$$

with integers satisfying $d = a+b$, $d \geq c \geq \max(a,b)$, $d \leq 2c$, $b \geq 1$ and $a-b = (-1)^\ell$. If $\ell \geq 2$, then $a \geq 1$ and $c \geq 2$. For $\ell \geq 0$ we have $C^\ell\mathbf{1} = (f_0, f_1, f_1)^{\mathsf{T}}$ with $0 < f_1 \leq f_0$.

Proof. For $\ell = 1$ we have $(d,c,a,b) = (1,1,0,1)$. Multiplication by C on the right gives $(d,c,a,b) \mapsto (d+2c,\ d+c,\ c+b,\ c+a)$, where the entry $(s,0)$ equals $c+a+b = d+c$ because $d = a+b$. All assertions follow by induction; for instance $2(d+c) - (d+2c) = d > 0$ and $(c+b) - (c+a) = -(a-b)$. For the last statement, $C(f_0,f_1,f_1)^{\mathsf{T}} = (f_0+2f_1, f_0+f_1, f_0+f_1)^{\mathsf{T}}$. □

4.1. **Charges.** Let $\mathbf{j}$ be a word of length $n \geq 1$. A site i is *neutral* if $j_i = 1$, a *point* if $j_i = 0$ and a *group* if $j_i \geq 2$. A point carries one *charge* and a group i carries $j_i - 1$ charges. We order all charges along the spine, the charges of a group being consecutive in a fixed order. A *coloring* χ assigns to each charge a color in $\{s,t\}$, and $\operatorname{wt}(\chi)$ is the product of all colors, a monomial in s,t. For a color κ let $\bar\kappa$ be the other color. Combinatorially, the charges of a group i are $j_i - 1$ of the pendant vertices of i that lie in the union of the two independent sets, each in exactly one of them, and the color of a charge records which one; the remaining factor v was removed in Lemma 3.4. The charge of a point is the spine vertex itself.

For a non-neutral site i and a coloring χ we define the *site matrix* $\Sigma_i(\chi)$ as follows. If i is a point whose charge has color κ, then $\Sigma_i(\chi) = E_{\kappa\kappa}$. If i is a group whose charges have colors $\kappa_1, \ldots, \kappa_a$, $a = j_i - 1$, then

$$\Sigma_i(\chi) = j_i\, E_{00} + \delta_i(\chi)\, E_{\bar\kappa_1\bar\kappa_a}, \tag{4.1}$$

where $\delta_i(\chi) = 1$ if $\kappa_1 = \cdots = \kappa_a$ and $\delta_i(\chi) = 0$ otherwise. Summing $\operatorname{wt}(\chi)\Sigma_i(\chi)$ over the colorings of the charges of site i gives $sE_{ss} + tE_{tt} = \Omega_0$ for a point, and $j_i u^a E_{00} + t^a E_{ss} + s^a E_{tt} = \Omega_{j_i}$ for a group. For a neutral site put $\Sigma_i = \Omega_1$, the identity matrix. Consequently

$$Z_{\mathbf{j}} = \sum_{\chi} \operatorname{wt}(\chi)\, \Phi(\chi), \qquad \Phi(\chi) = \mathbf{1}^{\mathsf{T}}\Sigma_1(\chi)\, C\, \Sigma_2(\chi)\, C \cdots C\, \Sigma_n(\chi)\, \mathbf{1}. \tag{4.2}$$

In the language of the three-state model, an occupied group is in the state opposite to the common color of its charges, and a point is in the state given by the color of its charge. The matrix $E_{\bar\kappa_1\bar\kappa_a}$ in (4.1) records this state through the first and

the last charge of the group; when $\delta_i(\chi) = 1$ the two coincide. This convention is needed for the sign expansion in Section 6.

4.2. **Color-symmetric matrices.** Let P be the permutation matrix exchanging the states s and t, and let $\mathcal{S}$ be the set of real 3×3 matrices K with $\mathsf{P}K\mathsf{P} = K$. These are exactly the matrices

$$K = \begin{pmatrix} d & p & p \\ q & a & b \\ q & b & a \end{pmatrix}, \tag{4.3}$$

and we write $g(K) = a - b$ and $w(K) = a + b$. The set $\mathcal{S}$ is closed under products and transposition, and contains C, E_{00} and the matrix

$$B = 6E_{00} - E_{st} - E_{ts} = \begin{pmatrix} 6 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & 0 \end{pmatrix}.$$

For $K \in \mathcal{S}$ of the form (4.3) the following facts are immediate.

- (S1) $K(0,1,-1)^{\mathsf{T}} = g(K)\,(0,1,-1)^{\mathsf{T}}$; hence g is multiplicative on $\mathcal{S}$, with $g(C) = -1$ and $g(B) = 1$.
- (S2) $K(A,H,H)^{\mathsf{T}} = (dA + 2pH,\ qA + wH,\ qA + wH)^{\mathsf{T}}$.
- (S3) For $\sigma, \sigma' \in \{s,t\}$ we have $K_{\sigma\sigma'} = b + g(K)\,[\sigma = \sigma']$, where $[\cdot]$ is the indicator.

We identify a vector $(A,H,H)^{\mathsf{T}}$ with the pair (A,H). By (S2),

$$C(A,H) = (A + 2H,\ A + H), \qquad CB(A,H) = (6A - 2H,\ 6A - H).$$

Let $\mathcal{R} = \{(A,H) : A > 0,\ 0 \le H \le 3A/2\}$.

Lemma 4.2. *The maps C and CB send $\mathcal{R}$ into $\mathcal{R}$, and they do not decrease either coordinate of a point of $\mathcal{R}$. Moreover CB multiplies the first coordinate by at least 3.*

Proof. Let $(A,H) \in \mathcal{R}$. For C, the image $(A + 2H, A + H)$ has positive first coordinate, $A + H \ge H \ge 0$, $A + 2H \ge A$, and $A + H \le \frac{3}{2}(A + 2H)$. For CB, the image satisfies $6A - 2H \ge 3A > 0$, $6A - H \ge H$ (as $H \le 3A$), and $6A - H \le \frac{3}{2}(6A - 2H)$, which is equivalent to $2H \le 3A$. □

5. Contraction of negatively signed pairs

5.1. **Splitting the sites with $j_i = 3$.** Let $\mathbf{j}$ be a word of length $n \ge 1$, let $T_3 = \{i : j_i = 3\}$ and $m_3 = |T_3|$. A site $i \in T_3$ is a group with two charges; for a coloring χ let κ_1, κ_2 be their colors and let $z_i(\chi) = [\kappa_1 = \kappa_2]$, so that $\delta_i(\chi) = z_i(\chi)$. Put

$$\Sigma_i^{\pm}(\chi) = 3E_{00} \pm E_{\bar\kappa_1 \bar\kappa_2}.$$

Checking the two cases $z_i(\chi) = 1$ and $z_i(\chi) = 0$ separately, one finds

$$\Sigma_i(\chi) = \tfrac{1}{2}\big(1 + z_i(\chi)\big)\Sigma_i^{+}(\chi) + \tfrac{1}{2}\big(1 - z_i(\chi)\big)\Sigma_i^{-}(\chi). \tag{5.1}$$

In the second term the two charges of i have opposite colors, and the occupied part of Σ_i^{-} carries the sign -1. These are the *negatively signed pairs*. Summing out their colors gives the following identity.

Proposition 5.1. *For every word* $\mathbf{j}$ *of length* $n \geq 1$,

$$Z_{\mathbf{j}} = 2^{-m_3} \sum_{S \subseteq T_3} v^{|S|}\, Y_S, \qquad Y_S = \sum_{\chi} \mathrm{wt}(\chi) \prod_{i \in T_3 \setminus S} \big(1 + z_i(\chi)\big)\, \Phi_S(\chi),$$

where χ *runs over the colorings of the charges of the sites not in* S*, and* $\Phi_S(\chi)$ *is obtained from* $\Phi(\chi)$ *in* (4.2) *by replacing* Σ_i *by* B *for* $i \in S$ *and by* Σ_i^+ *for* $i \in T_3 \setminus S$.

Proof. Write (5.1) as $\Sigma_i = \frac{1}{2}\big[(1+z_i)\Sigma_i^+ + (1-z_i)\Sigma_i^-\big]$, insert it for every $i \in T_3$ into (4.2), take out the factor 2^{-m_3} and expand; the set S records the sites at which the second term is chosen. For $i \in S$, the colors of the two charges of i occur only in $\mathrm{wt}(\chi)$, through the factor $\kappa_1\kappa_2$, in $1 - z_i(\chi)$ and in $\Sigma_i^-(\chi)$. The factor $1 - z_i$ restricts the sum to $\kappa_1 \neq \kappa_2$, where it equals 1 and $\kappa_1\kappa_2 = v$. Summing over $(\kappa_1, \kappa_2) \in \{(s,t),(t,s)\}$ gives

$$v\big(3E_{00} - E_{ts}\big) + v\big(3E_{00} - E_{st}\big) = v\, B,$$

which proves the identity. □

Example 5.2. For the word $\mathbf{j} = (3)$ we have $Z_{\mathbf{j}} = 3u^2 + t^2 + s^2$, and T_3 consists of the single site. For $S = \varnothing$, $\Phi_\varnothing(\chi) = \mathbf{1}^{\mathsf{T}}\Sigma_1^+(\chi)\mathbf{1} = 4$ for every coloring χ, so $Y_\varnothing = 4(2s^2 + 2t^2 + 2v)$. For $S = T_3$ we get $Y_{T_3} = \mathbf{1}^{\mathsf{T}} B \mathbf{1} = 4$. Proposition 5.1 gives $\frac{1}{2}\big(8s^2 + 8t^2 + 8v + 4v\big) = 4s^2 + 4t^2 + 6v$, which is indeed $3u^2 + s^2 + t^2 = 4h_2 + 2v$.

Fix $S \subseteq T_3$. The *surviving sites* are the non-neutral sites not in S; let them be $i_1 < \cdots < i_L$. Since C and B are symmetric matrices, grouping the factors of $\Phi_S(\chi)$ between consecutive surviving sites gives, for $L \geq 1$,

$$\Phi_S(\chi) = f_{\mathrm{L}}^{\mathsf{T}}\, \Xi_{i_1}(\chi)\, K_1\, \Xi_{i_2}(\chi)\, K_2 \cdots K_{L-1}\, \Xi_{i_L}(\chi)\, f_{\mathrm{R}}, \tag{5.2}$$

where $\Xi_i = \Sigma_i^+$ for $i \in T_3 \setminus S$ and $\Xi_i = \Sigma_i$ otherwise, each K_k belongs to

$$\mathcal{K} = \big\{ C^{\ell_0} B C^{\ell_1} B \cdots B C^{\ell_r} :\ r \geq 0,\ \ell_0, \ldots, \ell_r \geq 1 \big\},$$

with r the number of sites of S between i_k and i_{k+1}, and $f_{\mathrm{L}}, f_{\mathrm{R}}$ are *free vectors*, that is, vectors of the form

$$f = C^{\ell_0} B C^{\ell_1} B \cdots B C^{\ell_r} \mathbf{1}, \qquad r \geq 0,\ \ell_0, \ldots, \ell_{r-1} \geq 1,\ \ell_r \geq 0. \tag{5.3}$$

If $L = 0$, then Φ_S does not depend on χ and equals

$$\mathbf{1}^{\mathsf{T}} C^{\ell_0} B C^{\ell_1} B \cdots B C^{\ell_r} \mathbf{1}, \qquad r = |S|,\ \ell_0, \ell_r \geq 0,\ \ell_1, \ldots, \ell_{r-1} \geq 1. \tag{5.4}$$

All matrices in $\mathcal{K}$ belong to $\mathcal{S}$, and $\mathcal{K}$ is closed under transposition.

5.2. **Bounds for contracted kernels.**

Lemma 5.3. *Let* $K = C^{\ell_0} B C^{\ell_1} \cdots B C^{\ell_r} \in \mathcal{K}$ *with* $r \geq 1$*, written as in* (4.3)*, and let* $g = g(K)$*,* $w = w(K)$*. Then the entries of* K *are integers and*

(a) $g \in \{-1, 1\}$*;*
(b) $p \leq \frac{3}{2}d$*,* $q \leq \frac{3}{2}d$*,* $w \leq 3p$ *and* $w \leq 3q$*;*
(c) $p \geq 5$*,* $q \geq 5$ *and* $w \geq 11$*;*
(d) $\Delta(K) := 8d - 4p - 4q + w \geq 3$.

Part (a) also holds for $r = 0$.

Proof. Part (a) follows from (S1), and integrality is clear. We use the factorization

$$K = C^{\ell_0-1}(CB)\,C^{\ell_1-1}(CB)\cdots C^{\ell_{r-1}-1}(CB)\,C^{\ell_r}$$

and Lemma 4.2. Let $e_0 = (1,0,0)^{\mathsf{T}}$, which corresponds to $(1,0) \in \mathcal{R}$. By (S2), $Ke_0 = (d,q,q)^{\mathsf{T}}$, so $(d,q) \in \mathcal{R}$ and $q \le \frac{3}{2}d$. Applying this to $K^{\mathsf{T}} \in \mathcal{K}$, whose first column is $(d,p,p)^{\mathsf{T}}$, gives $p \le \frac{3}{2}d$. Next, $C(0,1,1)^{\mathsf{T}} = (2,1,1)^{\mathsf{T}} \in \mathcal{R}$, and since $\ell_r \ge 1$ the vector $K(0,1,1)^{\mathsf{T}} = (2p,w,w)^{\mathsf{T}}$ lies in $\mathcal{R}$; thus $w \le 3p$, and by transposition $w \le 3q$. This proves (b).

For (c), let (d',c',a',b') be the entries of C^{ℓ_r} as in Lemma 4.1. Then $C^{\ell_r}e_0 = (d',c')$ with $1 \le c' \le d'$, and CB maps it to $(6d'-2c', 6d'-c')$, whose coordinates are at least 4 and 5. By the monotonicity in Lemma 4.2, $q \ge 5$, and by transposition $p \ge 5$. Similarly $C^{\ell_r}(0,1,1)^{\mathsf{T}} = (2c',d')$ by (S2) and $d' = a'+b'$, and CB maps it to $(12c'-2d', 12c'-d')$. If $\ell_r = 1$ the second coordinate is 11; if $\ell_r \ge 2$ it is at least $10c' \ge 20$, since $d' \le 2c'$ and $c' \ge 2$. Monotonicity gives $w \ge 11$.

For (d), let $\mathbf{k} = (4,-1,-1)^{\mathsf{T}}$. A direct computation gives $\mathbf{k}^{\mathsf{T}}K\mathbf{k} = 16d - 8p - 8q + 2w = 2\Delta(K)$. By (S2), $C\mathbf{k}$ corresponds to $(2,3) \in \mathcal{R}$. Applying the remaining factors of K, at least one of which is CB, we obtain $K\mathbf{k} = (A,H,H)^{\mathsf{T}}$ with $(A,H) \in \mathcal{R}$ and $A \ge 3 \cdot 2 = 6$. Hence $2\Delta(K) = 4A - 2H \ge A \ge 6$. □

Lemma 5.4. *Every free vector* (5.3) *has the form* $(f_0,f_1,f_1)^{\mathsf{T}}$ *with* $f_0 > 0$ *and* $0 < f_1 \le \frac{3}{2}f_0$. *If* $r = 0$, *then* $f_1 \le f_0$.

Proof. The vector $\mathbf{1}$ corresponds to $(1,1) \in \mathcal{R}$. Write

$$f = C^{\ell_0-1}(CB)\cdots C^{\ell_{r-1}-1}(CB)\,C^{\ell_r}\mathbf{1}$$

and apply Lemma 4.2; the case $r = 0$ is Lemma 4.1. The second coordinate stays positive, since C and CB map (A,H) with $A > 0$ and $0 < H \le \frac{3}{2}A$ to points with second coordinate $A + H > 0$ and $6A - H > 0$, respectively. □

Lemma 5.5. *The scalar* (5.4) *is positive.*

Proof. For $r = 0$ it equals $\mathbf{1}^{\mathsf{T}}C^{\ell_0}\mathbf{1} > 0$. For $r \ge 1$ write it as $(C^{\ell_0}\mathbf{1})^{\mathsf{T}}B\,\mathbf{y}$ with $\mathbf{y} = C^{\ell_1}B\cdots BC^{\ell_r}\mathbf{1}$, a free vector with $r-1$ factors B. By Lemmas 4.1 and 5.4, $C^{\ell_0}\mathbf{1} = (f_0,f_1,f_1)^{\mathsf{T}}$ with $0 < f_1 \le f_0$ and $\mathbf{y} = (A,H,H)^{\mathsf{T}}$ with $A > 0$ and $0 < H \le \frac{3}{2}A$. Hence the scalar equals $6f_0A - 2f_1H \ge 6f_0A - 3f_0A > 0$. □

6. Sign expansion and the boundary-ratio interval

6.1. **Equality variables.** Fix a word $\mathbf{j}$ of length $n \ge 1$ and $S \subseteq T_3$ with $L \ge 1$ surviving sites. Let $\gamma_1,\dots,\gamma_N$ be the charges of the surviving sites in their order; $N \ge 1$. Each consecutive pair $\{\gamma_e,\gamma_{e+1}\}$ is an *edge*. It is *internal* if both charges belong to the same site and a *junction* otherwise; there is exactly one junction e_k between the sites i_k and i_{k+1}. The internal edges of the sites in $T_3 \setminus S$ are called *good*; all other edges, namely the junctions and the internal edges of groups with $j_i \ge 4$, are *active*. Let m' be the number of active edges. For a coloring χ let $z_e(\chi) = 1$ if γ_e and γ_{e+1} have the same color and $z_e(\chi) = 0$ otherwise; for $i \in T_3 \setminus S$, $z_i(\chi)$ is the value at its good edge.

Each surviving site i has a set of *options*: a point has the single option $\bullet$, and a group has the options $\circ$ (vacant) and $\bullet$ (occupied). The multipliers of the options are $\mu_i(\circ) = j_i$, $\mu_i(\bullet) = 1$ if i is a point or $j_i \in \{2,3\}$, and

$$\mu_i(\bullet) = \prod_{e \text{ internal to } i} z_e \qquad \text{if } j_i \ge 4.$$

For $1 \le k < L$ write K_k as in (4.3) with entries d_k, p_k, q_k, a_k, b_k and $g_k = a_k - b_k$, and define the junction weights

$$\tau_k(\circ,\circ) = d_k, \qquad \tau_k(\circ,\bullet) = p_k, \qquad \tau_k(\bullet,\circ) = q_k,$$
$$\tau_k(\bullet,\bullet) = \begin{cases} b_k + g_k z_{e_k}, & \text{if } i_k \text{ and } i_{k+1} \text{ are of the same kind,} \\ a_k - g_k z_{e_k}, & \text{otherwise,} \end{cases}$$

where the two kinds are points and groups. For the free vectors $f_{\mathrm{L}} = (f_0^{\mathrm{L}}, f_1^{\mathrm{L}}, f_1^{\mathrm{L}})^{\mathsf{T}}$ and $f_{\mathrm{R}} = (f_0^{\mathrm{R}}, f_1^{\mathrm{R}}, f_1^{\mathrm{R}})^{\mathsf{T}}$ put $\varphi_{\mathrm{L}}(\circ) = f_0^{\mathrm{L}}$, $\varphi_{\mathrm{L}}(\bullet) = f_1^{\mathrm{L}}$, and similarly for φ_{R}. Define

$$F_S(z) = \sum_{o} \varphi_{\mathrm{L}}(o_1) \prod_{k=1}^{L} \mu_{i_k}(o_k) \prod_{k=1}^{L-1} \tau_k(o_k, o_{k+1})\, \varphi_{\mathrm{R}}(o_L), \tag{6.1}$$

the sum running over all choices of options $o = (o_1, \dots, o_L)$. This is a polynomial in the variables z_e of the active edges, affine in each of them: a junction variable occurs only in one factor τ_k, which is affine, and an internal variable occurs only in $\mu_i(\bullet)$, as one factor of a product of distinct variables.

Lemma 6.1. *For every coloring χ of the surviving charges, $\Phi_S(\chi) = F_S\big(z(\chi)\big)$.*

Proof. In (5.2), a point i with color κ contributes $E_{\kappa\kappa}$. A group i whose first and last charges have colors κ, κ' contributes $j_i E_{00} + \mu E_{\bar\kappa\bar\kappa'}$, where $\mu = \delta_i(\chi)$ if $j_i \neq 3$ and $\mu = 1$ if $i \in T_3 \setminus S$. If $j_i = 2$, then $\delta_i(\chi) = 1$; if $j_i \ge 4$, then $\delta_i(\chi)$ is the product of $z_e(\chi)$ over the internal edges of i. Thus $\mu = \mu_i(\bullet)(z(\chi))$. Expanding (5.2) along these decompositions yields a sum over options. The vacant option uses the state 0 on both sides of the site; the occupied option uses the state κ (point) or $\bar\kappa$ (group) determined by the first charge on the left, and the state determined by the last charge on the right. The entries of f_{L}, f_{R} and K_k indexed by the state 0 are f_0, d_k, p_k, q_k. By (S3), the entry of K_k between occupied states σ, σ' is $b_k + g_k[\sigma = \sigma']$. If i_k and i_{k+1} are of the same kind, then $[\sigma = \sigma'] = z_{e_k}(\chi)$; otherwise exactly one of the two states is the color of a charge and the other is opposite to a color, so $[\sigma = \sigma'] = 1 - z_{e_k}(\chi)$ and $b_k + g_k(1 - z_{e_k}) = a_k - g_k z_{e_k}$. This is (6.1) at $z = z(\chi)$. □

Lemma 6.2. *If F is a polynomial in $z_1, \dots, z_m$ that is affine in each variable, then*

$$F(z) = 2^{-m} \sum_{\varepsilon \in \{-1,1\}^m} F(\varepsilon) \prod_{e=1}^{m} (1 + \varepsilon_e z_e).$$

Proof. For $m = 1$ and $F = F_0 + F_1 z$ the right side is $\frac{1}{2}(F_0 + F_1)(1 + z) + \frac{1}{2}(F_0 - F_1)(1 - z) = F(z)$. The general case follows by applying this identity to one variable at a time. □

Proposition 6.3. *If $L \ge 1$, then*

$$Y_S = 2^{-m'} \sum_{\varepsilon} F_S(\varepsilon)\, X_N\big(\boldsymbol{\lambda}(\varepsilon)\big),$$

where ε runs over the sign vectors on the active edges, $\lambda_e(\varepsilon) = 1 + \varepsilon_e$ for an active edge e, and $\lambda_e(\varepsilon) = 2$ for a good edge e.

Proof. By Lemmas 6.1 and 6.2, $\Phi_S(\chi) = 2^{-m'} \sum_\varepsilon F_S(\varepsilon) \prod_{e \text{ active}} (1 + \varepsilon_e z_e(\chi))$. For colors κ, κ' of the endpoints of e we have $1 + \varepsilon_e z_e(\chi) = W_{1+\varepsilon_e}(\kappa, \kappa')$ and, for a good edge, $1 + z_e(\chi) = W_2(\kappa, \kappa')$. Substituting into the definition of Y_S and summing over χ gives $X_N(\boldsymbol{\lambda}(\varepsilon))$. □

Remark 6.4. The coefficients produced by a sign expansion need not be nonnegative, and this is the reason for Proposition 5.1. If the internal edges of the sites in T_3 are treated as active edges and no contraction is performed, the analogue of F_S for the word $\mathbf{j} = (0,3,3,3,3,3,0)$, with sign -1 on all five internal edges and sign $+1$ on all junctions, equals -9. Such a group has occupied multiplier -1 and $j_i = 3$, so the implication (6.2) in the proof of Proposition 6.7 below fails for it: $-x/3$ can be as small as $-2/3$.

6.2. **Nonnegativity of the coefficients.** For $K \in \mathcal{K}$ written as in (4.3), with $g = g(K)$ and $\varepsilon \in \{-1, 1\}$, put

$$\theta_\varepsilon = b + g\varepsilon, \qquad \theta'_\varepsilon = a - g\varepsilon.$$

By (6.1), θ_ε is the occupied junction weight between sites of the same kind and θ'_ε the one between sites of different kinds.

Lemma 6.5. *For every $K \in \mathcal{K}$ and $\varepsilon \in \{-1, 1\}$:*

(K1) $p, q > 0$ *and* $2d > \max(p, q)$*;*
(K2) $0 \le \theta_\varepsilon \le 2\min(p, q)$*;*
(K3) $-\min(p, q) \le \theta'_\varepsilon \le 2\min(p, q)$*;*
(K4) $4d - 2p - 2q + \theta_\varepsilon \ge 0$.

Proof. If $K = C^\ell$, Lemma 4.1 gives $p = q = c \ge 1$, $d \ge c$, $b \ge 1$, $0 \le a \le c$, $b \le c$ and $|g| = 1$. Hence $\theta_\varepsilon \in \{b-1, b+1\} \subseteq [0, c+1] \subseteq [0, 2c]$ and $\theta'_\varepsilon \in \{a-1, a+1\} \subseteq [-1, c+1] \subseteq [-c, 2c]$; (K1) and (K4) are clear.

If K contains B, let $w = w(K)$. Since $a = (w+g)/2$, $b = (w-g)/2$ and $g = \pm 1$ by Lemma 5.3(a), both θ_ε and θ'_ε belong to $\{(w \pm g)/2, (w \pm 3g)/2\} \subseteq [(w-3)/2, (w+3)/2]$. By Lemma 5.3(c), $(w-3)/2 \ge 4$. With $m = \min(p, q) \ge 5$, Lemma 5.3(b) gives $(w+3)/2 \le (3m+3)/2 \le 2m$. This proves (K2) and (K3). For (K4), $4d-2p-2q+\theta_\varepsilon \ge 4d-2p-2q+(w-3)/2 = (\Delta(K)-3)/2 \ge 0$ by Lemma 5.3(d). Finally (K1) follows from Lemma 5.3(b),(c), since $\max(p, q) \le \frac{3}{2}d < 2d$. □

Lemma 6.6. *Let $K \in \mathcal{K}$, $\varepsilon \in \{-1, 1\}$ and $\zeta \in [-\frac{1}{2}, 1]$. Then $d + q\zeta > 0$ and*

$$0 \le R(\zeta) := \frac{p + \theta_\varepsilon \zeta}{d + q\zeta} \le 2.$$

Proof. By (K1), $d + q\zeta \ge d - q/2 > 0$. The numerator $p + \theta_\varepsilon\zeta$ is affine in ζ; at $\zeta = -\frac{1}{2}$ it is $p - \theta_\varepsilon/2 \ge 0$ by (K2), and at $\zeta = 1$ it is $p + \theta_\varepsilon \ge 0$. Similarly $2(d+q\zeta) - (p+\theta_\varepsilon\zeta)$ equals $\frac{1}{2}(4d-2p-2q+\theta_\varepsilon) \ge 0$ at $\zeta = -\frac{1}{2}$, by (K4), and equals $(2d-p) + (2q-\theta_\varepsilon) > 0$ at $\zeta = 1$, by (K1) and (K2). □

Proposition 6.7. *For every $S \subseteq T_3$ with $L \ge 1$ and every sign vector ε on the active edges, $F_S(\varepsilon) \ge 0$.*

Proof. For a group i let $\eta_i = \mu_i(\bullet)(\varepsilon)$. Then $\eta_i = 1$ if $j_i \in \{2, 3\}$ and $\eta_i \in \{-1, 1\}$ if $j_i \ge 4$. In each of these cases,

$$x \in [-1, 2] \quad \Longrightarrow \quad \eta_i x / j_i \in [-\tfrac{1}{2}, 1], \tag{6.2}$$

since $x/2 \in [-\frac{1}{2}, 1]$, $x/3 \in [-\frac{1}{3}, \frac{2}{3}]$ and $|x|/j_i \le \frac{1}{2}$ for $j_i \ge 4$.

Since a point has a single option, the sum (6.1) factorizes along the points. More precisely, $F_S(\varepsilon)$ is the product of the following factors: f_1^{L} if i_1 is a point and f_1^{R} if i_L is a point; θ_ε (for the kernel K_k and the sign of e_k) for every pair i_k, i_{k+1} of consecutive points; and the value of every *run*, that is, every maximal

sequence of consecutive surviving sites that are groups. The first two kinds of factors are nonnegative by Lemma 5.4 and (K2). It remains to show that every run has nonnegative value.

Consider a run $i_{\underline{k}}, \dots, i_{\overline{k}}$. Its left boundary is the free end if $\underline{k} = 1$ and the point $i_{\underline{k}-1}$ otherwise. In the first case put $(\xi_\circ, \xi_\bullet) = (f_0^{\mathrm{L}}, f_1^{\mathrm{L}})$. In the second case put $(\xi_\circ, \xi_\bullet) = (q_{\underline{k}-1}, \theta'_{\underline{k}-1})$; these are the junction weights $\tau_{\underline{k}-1}(\bullet, \circ)$ and $\tau_{\underline{k}-1}(\bullet, \bullet)$ from a point to a group, where $\theta'_{\underline{k}-1}$ is formed with $K_{\underline{k}-1}$ and the sign of $e_{\underline{k}-1}$. Similarly, put $(\psi_\circ, \psi_\bullet) = (f_0^{\mathrm{R}}, f_1^{\mathrm{R}})$ if $\overline{k} = L$, and $(\psi_\circ, \psi_\bullet) = (p_{\overline{k}}, \theta'_{\overline{k}})$ if $i_{\overline{k}+1}$ is a point. By Lemma 5.4 and (K1), (K3), we have $\xi_\circ, \psi_\circ > 0$, and the boundary ratios $\rho_{\mathrm{L}} = \xi_\bullet/\xi_\circ$ and $\rho_{\mathrm{R}} = \psi_\bullet/\psi_\circ$ lie in $[-1, 2]$. The value of the run is

$$V = \sum_{o_{\underline{k}}, \dots, o_{\overline{k}}} \xi_{o_{\underline{k}}} \prod_{k=\underline{k}}^{\overline{k}} \mu_{i_k}(o_k) \prod_{k=\underline{k}}^{\overline{k}-1} \tau_k(o_k, o_{k+1})\, \psi_{o_{\overline{k}}},$$

evaluated at ε. Define $(A_{\underline{k}}, H_{\underline{k}}) = (j_{i_{\underline{k}}}\xi_\circ,\ \eta_{i_{\underline{k}}}\xi_\bullet)$ and, for $\underline{k} \le k < \overline{k}$,

$$A_{k+1} = j_{i_{k+1}}\big(A_k d_k + H_k q_k\big), \qquad H_{k+1} = \eta_{i_{k+1}}\big(A_k p_k + H_k \theta_k\big),$$

where θ_k is formed with K_k and the sign of e_k; here two consecutive groups are of the same kind. Summing over the options one site at a time shows that $V = A_{\overline{k}}\psi_\circ + H_{\overline{k}}\psi_\bullet$.

We claim that $A_k > 0$ and $\zeta_k := H_k/A_k \in [-\frac{1}{2}, 1]$ for $\underline{k} \le k \le \overline{k}$. For $k = \underline{k}$ this holds because $\zeta_{\underline{k}} = \eta_{i_{\underline{k}}}\rho_{\mathrm{L}}/j_{i_{\underline{k}}}$ and $\rho_{\mathrm{L}} \in [-1, 2]$, by (6.2). If it holds for k, then $A_{k+1} = j_{i_{k+1}} A_k(d_k + q_k\zeta_k) > 0$ by Lemma 6.6, and $\zeta_{k+1} = \eta_{i_{k+1}} R_k(\zeta_k)/j_{i_{k+1}}$, where $R_k(\zeta_k) \in [0, 2]$ is the ratio of Lemma 6.6 for K_k; thus $\zeta_{k+1} \in [-\frac{1}{2}, 1]$ by (6.2). Consequently

$$V = A_{\overline{k}}\,\psi_\circ\big(1 + \rho_{\mathrm{R}}\zeta_{\overline{k}}\big) \ge 0,$$

because $\rho_{\mathrm{R}} \in [-1, 2]$ and $\zeta_{\overline{k}} \in [-\frac{1}{2}, 1]$ give $\rho_{\mathrm{R}}\zeta_{\overline{k}} \ge -1$; the bilinear function $\rho\zeta$ attains its minimum -1 over this rectangle at the corners $(-1, 1)$ and $(2, -\frac{1}{2})$. When the surviving sites contain no point, the whole sequence $i_1, \dots, i_L$ is a single run with two free ends, and the same argument applies. □

6.3. **Proofs of Theorems 3.5 and 1.1.**

Proof of Theorem 3.5. For the empty word $Z_{\mathbf{j}} = 1$. Let $n \ge 1$. By Proposition 5.1, $Z_{\mathbf{j}} = 2^{-m_3}\sum_S v^{|S|} Y_S$. If $L = 0$, there are no surviving charges and $Y_S = \Phi_S$ is the positive constant (5.4), by Lemma 5.5; this includes words consisting only of neutral sites. If $L \ge 1$, Propositions 6.3 and 6.7 express Y_S as a nonnegative combination of free Ising chains $X_N(\boldsymbol{\lambda})$ with $\boldsymbol{\lambda} \in \{0, 2\}^{N-1}$, which lie in Λ^+ by Lemma 2.4. Since $v^{|S|} \in \Lambda^+$, Lemma 2.1(c) gives $Z_{\mathbf{j}} \in \Lambda^+$. □

Proof of Theorem 1.1. By Theorem 3.5 and Lemmas 3.4 and 3.3, $A_{\mathbf{j}} \in \Lambda^+$ for every word $\mathbf{j}$. By Lemma 3.2, $I(T; s)I(T; t)$ is a nonnegative combination of the polynomials $A_{\mathbf{j}}$, so it lies in Λ^+. Lemma 2.2 applied to $P = I(T; \cdot)$ gives $i_k(T)^2 \ge i_{k-1}(T)i_{k+1}(T)$ for all $k \ge 1$. □

Remark 6.8. The proof shows that $I(T; s)I(T; t)$ is Schur positive in two variables, and Lemma 2.1(a) then also gives $i_a(T)i_b(T) \ge i_{a+1}(T)i_{b-1}(T)$ for all $a \ge b \ge 1$. For positive sequences this strong log-concavity is equivalent to log-concavity; cf. [10, Theorem 1.3].

Remark 6.9. Theorem 1.1 does not extend to real-rootedness. The star $K_{1,3} = T(3)$ has independence polynomial $1 + 4x + 3x^2 + x^3$, whose discriminant is -31, so two of its roots are not real.

Remark 6.10. The theorem concerns the ordinary count of independent sets and does not extend to arbitrary positive vertex weights. If the center of $K_{1,3}$ receives weight 7 and every leaf weight 1, the weighted independence polynomial $\sum_I \big(\prod_{y\in I} w_y\big)x^{|I|}$ equals $1 + 10x + 3x^2 + x^3$, and $3^2 < 10 \cdot 1$.

Remark 6.11. By Theorem 1.1, none of the trees in [4, 7, 8] whose independence sequences are not log-concave is a caterpillar; thus the theorem does not extend to all trees.

Use of AI-assisted tools

AI-assisted tools were used in the research and preparation of this manuscript.

References

[1] Y. Alavi, P. J. Malde, A. J. Schwenk, and P. Erdős, *The vertex independence sequence of a graph is not constrained*, Congr. Numer. **58** (1987), 15–23.

[2] P. Bahls, B. Ethridge, and L. Szabo, *Unimodality of the independence polynomials of non-regular caterpillars*, Australas. J. Combin. **71** (2018), no. 1, 104–112.

[3] E. X. Fang, J. Lu, E. Nevo, Y. Yao, and H. Zheng, *Unimodality of independence polynomials for sufficiently large forests*, preprint (2026), arXiv:2609.20961v1.

[4] D. Galvin, *Trees with non log-concave independent set sequences*, preprint (2026), arXiv:2502.10654v2.

[5] D. Galvin and J. Hilyard, *The independent set sequence of some families of trees*, Australas. J. Combin. **70** (2018), no. 2, 236–252.

[6] I. Gutman and F. Harary, *Generalizations of the matching polynomial*, Utilitas Math. **24** (1983), 97–106.

[7] O. Kadrawi and V. E. Levit, *The independence polynomial of trees is not always log-concave starting from order* 26, preprint (2023), arXiv:2305.01784.

[8] O. Kadrawi, V. E. Levit, R. Yosef, and M. Mizrachi, *On computing of independence polynomials of trees*, in: Recent Research in Polynomials (F. Özger, ed.), IntechOpen, 2023.

[9] V. E. Levit and E. Mandrescu, *Very well-covered graphs with log-concave independence polynomials*, Carpathian J. Math. **20** (2004), no. 1, 73–80.

[10] E. Y. H. Li, G. M. X. Li, A. L. B. Yang, and Z.-X. Zhang, *A symmetric function approach to log-concavity of independence polynomials*, Selecta Math. (N.S.) **32** (2026), Paper No. 87. Theorem numbers cited here refer to arXiv:2501.04245v1.

[11] W. Li, K. Vallier, and T. Zhang, *Unimodality of forest independence polynomials*, provisional manuscript (2026), arXiv:2610.07943v1.

[12] R. P. Stanley, *Graph colorings and related symmetric functions: ideas and applications*, Discrete Math. **193** (1998), 267–286.

[13] Y. Wang and B.-X. Zhu, *On the unimodality of independence polynomials of some graphs*, European J. Combin. **32** (2011), 10–20.

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